

C lib.c

C libabry.c

≡ issue.txt

G+ day68.c

C day69.c

C day70.c

C day71.c

C dyn.c > ...

```
1  #include <stdio.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  int main() {
10     struct Node *head = NULL, *temp = NULL, *newNode = NULL;
11     int n, value;
12
13     printf("Enter number of nodes: ");
14     scanf("%d", &n);
15
16     for (int i = 0; i < n; i++) {
17         printf("Enter value for node %d: ", i + 1);
18         scanf("%d", &value);
19
20         newNode = (struct Node*)malloc(sizeof(struct Node));
21         newNode->data = value;
22         newNode->next = NULL;
23         if (head == NULL)
24             head = newNode;
25         else
26             temp->next = newNode;
27         temp = newNode;
28     }
29
30     printf("Linked List: ");
31     temp = head;
32     while (temp != NULL) {
33         printf("%d -> ", temp->data);
34         temp = temp->next;
35     }
36     printf("NULL\n");
37 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\ACER\Desktop\jss> gcc dyn.c

PS C:\Users\ACER\Desktop\jss> ./a.exe

Enter number of nodes: 9

Enter value for node 1: 5

Enter value for node 2: 7

Enter value for node 3: 5

Enter value for node 4: 4

Linked List: 5 -> 7 -> 5 -> 4 -> 3 -> 4 -> 9 -> 10 -> 4 -> NULL

Linked List: 5 -> 7 -> 5 -> 4 -> 3 -> 4 -> 9 -> 10 -> 4 -> NULL

PS C:\Users\ACER\Desktop\jss> □

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Experiment 10 : Dynamic Memory Allocation

1. WAP to create a simple linked list in C using pointer and structure.

```
#include <stdio.h>
#include <stdlib.h>
```

```
struct Node {
    int data;
    struct Node * next;
};
```

```
int main() {
    struct Node * head = NULL, * temp = NULL, * new Node = NULL;
    int n, value;
```

```
printf("Enter number of nodes: ");
scanf("%d", &n);
```

```
for(int i=0; i<n; i++) {
    printf("Enter value for node %d: ", i+1);
    scanf("%d", &value);
```

```
new Node = (struct Node *) malloc (size of (struct Node));
new Node -> data = value;
new Node -> next = NULL;
```

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dyn.c > ...

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node* next;
7  };
8
9  void printList(struct Node* head) {
10     struct Node* temp = head;
11     while (temp != NULL) {
12         printf("%d -> ", temp->data);
13         temp = temp->next;
14     }
15     printf("NULL\n");
16 }
17
18 int main() {
19     struct Node *head = NULL, *temp = NULL, *newNode = NULL;
20     int n, value, pos;

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\ACER\Desktop\jss> gcc dyn.c

PS C:\Users\ACER\Desktop\jss> ./a.exe

Enter number of nodes: 2

Enter value for node 1: 2

Enter value for node 2: 4

Original Linked List: 2 -> 4 -> NULL

Enter position to insert (starting from 1):

```
if (head == NULL) {  
    head = newNode;  
    temp = newNode;  
} else {  
    temp->next = newNode;  
    temp = newNode;  
}  
  
printf("Linked list:");  
temp = head;  
while (temp != NULL) {  
    printf("%d -> ", temp->data);  
    temp = temp->next;  
}  
  
printf("NULL");  
return 0;  
}
```

2. Insert an item in the Middle of the linked list

```
#include <stdio.h>  
#include <stdlib.h>
```

```
struct Node {  
    int data;
```

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```
struct Node * next;  
};
```

```
void printlist(struct Node * head) {  
    struct Node * temp = head;  
    while (temp != NULL) {  
        printf("%d →", temp → data);  
        temp = temp → next;  
    }  
    printf("NULL\n");  
}
```

```
int main() {  
    struct Node * head = NULL, * temp = NULL, * newNode = NULL;  
    int n, value, pos;
```

```
    printf("Enter number of nodes: ");  
    scanf("%d", &n);
```

```
    for (int i = 0; i < n; i++) {  
        printf("Enter value for node %d: ", i + 1);  
        scanf("%d", &value);
```

```
        newNode = (struct Node *) malloc (size of (struct Node));  
        newNode → data = value;
```

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newNode → next = NULL;

```
if (head == NULL) {  
    head = temp = newNode;  
} else {  
    temp → next = newNode;  
    temp = newNode;  
}
```

```
printf("Original linked list");  
printlist(head);
```

```
printf("Enter position to insert (starting from 1): ");  
scanf("%d", &pos);
```

```
printf("Enter value to insert: ");  
scanf("%d", &value);
```

```
newNode = (struct Node *) malloc(sizeof(struct Node));  
newNode → data = value;  
newNode → next = NULL;
```

```
printf("linked list:");  
temp = head;  
while (temp != NULL) {
```

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```
newNode → next = NULL;
```

```
if (head == NULL) {  
    head = temp = newNode;  
} else {  
    temp → next = newNode;  
    temp = newNode;  
}
```

```
{  
    printf("Original linked list");  
    printlist(head);
```

```
printf("Enter position to insert (starting from 1): ");  
scanf("%d", &pos);
```

```
printf("Enter value to insert: ");  
scanf("%d", &value);
```

```
newNode = (struct Node *) malloc(sizeof(struct Node));  
newNode → data = value;  
newNode → next = NULL;
```

```
printf("linked list:");  
temp = head;  
while (temp != NULL) {
```

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```
printf("linked list:");  
temp = head;  
while (temp != NULL) {  
    printf("%d → ", temp->data);  
    temp = temp->next;  
}  
printf("NULL");  
return 0;  
}
```

Experiment 11: Bitwise Operator

1. WAP to apply bitwise OR, AND and NOT operator on bit level.

```
#include <stdio.h>
```

```
int main() {  
    int a, b;
```

```
    printf("Enter 2 numbers:");  
    scanf("%d %d", &a, &b);
```

```
    printf("In Bitwise AND (a & b) = %d", a & b);  
    printf("In Bitwise OR (a | b) = %d", a | b);  
    printf("In Bitwise NOT (~a) = %d", ~a);
```

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